Dongwook Kwon

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Summary

Highly accomplished AI Research Engineer specializing in Large Language Model (LLM) optimization, Multi-Agent Systems (MAS), and AI-driven anomaly detection for critical infrastructure. Demonstrated research excellence through co-first authored publications in top-tier peer-reviewed venues, including an Oral presentation at EMNLP 2025 and a paper at ACL 2026 (both Industry Tracks), developed in collaboration with LG Electronics, and a first-authored journal article in Data Mining and Knowledge Discovery (Springer Nature) on anomaly detection for industrial control systems in power generation. Inventor of multiple registered patents in network and medical device anomaly detection. A core engineer who engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company. With foundational research experience in nuclear cybersecurity and medical cyber-physical systems, poised to significantly advance the United States' technological edge in critical and emerging fields by engineering scalable, reliable, and secure autonomous AI architectures.

Patents

Registered

Nov 2026 **Categorical Data of Networks Based Network Anomaly Detection Apparatus and Method Thereof**, KR 10-3043922

Apr 2026 **Method for Implantable Medical Device to Detect Anomaly in Real Time**, KR 10-2961984

Aug 2025 **GPT API-Based Network Anomaly Detection**, KR 10-2848814

Pending Applications

Oct 2025 **System for Anomaly Detection Using a Collaborative Multi-AI Determination Module Based on LLM**, KR 10-2025-0160217

Publications

Journal Articles

D. Kwon, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," *Data Mining and Knowledge Discovery*, vol. 40, art. no. 102, 2026.

Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.

Conference Papers

R. Chang†, **D. Kwon**†, J. Lee†, and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL), Industry Track (Poster)*, pp. 1329–1340, 2026. †Equal contribution.

J. Lee†, R. Chang†, **D. Kwon**†, H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, pp. 1522–1532, 2025. †Equal contribution.

D. Kwon, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. **Best Poster Award — Silver.**

D. Kwon, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. **Outstanding Paper Award.**

J. Yang, J. Kim, J. Lee, H. Ryu, **D. Kwon**, S. Yeo, P. Kim, Y. Kim, J. Lim, H. Yoon, and C. Park, "Metaverse: Design of the Car Price Prediction Model Through a Machine-learning Approach," in *Proc. IEEE Int. Conf. Metaverse Computing, Networking and Applications (MetaCom)*, pp. 734–737, 2023.

Research & Professional Experience

AI Engineer, Intelligent Agent Technology Team Dec. 2025 – Present
Suresoft Technologies *Seongnam, Korea*

- Developing an AI validation system utilizing AI agent–based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company.

Research Assistant Aug. 2021 – Dec. 2025
Bio-Computing & ML Lab, Kwangwoon University *Seoul, Korea*

- Researched deep-learning algorithms for anomaly detection in complex time-series data, focusing on critical cyber-physical systems including nuclear power generation environments.
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.

Visiting Graduate Researcher Jan. 2025 – Jul. 2025
University of Toronto *Toronto, ON, Canada*

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

AI Development Project Participant, Summer Program Jul. 2022 – Aug. 2022
Qualcomm Institute, UC San Diego *San Diego, CA, USA*

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

Education

Kwangwoon University Seoul, Korea
M.S. in Computer Engineering, GPA 4.33 / 4.5 (98.1%) *Mar. 2024 – Feb. 2026*

University of Toronto Toronto, ON, Canada
Graduate Research Program, GPA 3.68 / 4.0 *Jan. 2025 – Jul. 2025*

Kwangwoon University Seoul, Korea
B.S. in Electronics and Communications Engineering, GPA 3.41 / 4.5 (88.1%) *Mar. 2019 – Feb. 2024*

Honors & Awards

Major Awards

Aug. 2022 **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego

Dec. 2021 **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries

Academic Recognition

Nov. 2025 **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers

Nov. 2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024)

Nov. 2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering